

JAVASCRIPT PRACTICAL LAB – 3

Practice: Variables, Data Types & Operators

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Course: BCA

Semester: 5th Semester

Subject: JavaScript

Part 0 — Quick Recap Warm-Up

```
> console.log(typeof (10 + "5"));
console.log(5 === "5");

let numberX = 10;
numberX += 5;
console.log(numberX);

console.log(true && false || true);

console.log(10 % 3);
```

string	VM458:1
false	VM458:2
15	VM458:6
true	VM458:8
1	VM458:10

```
< undefined
>
```

Part 1 — Operator Precedence: Level Up

Task 1.1 — Trace It Yourself:

Task 1.2 — Comparison Chaining Gotcha:

Task 1.3

```
> let val = 4 + 3 * 2 ** 2 - 6 / 3;
console.log(val);

14 VM462:2

< undefined

> console.log(1 < 2 < 3);
console.log(3 > 2 > 1);

true VM466:1
false VM466:2

< undefined

> let answer = (5 + 3 * 2) > 10 && 20 / 2 === 10;

console.log(answer);

true VM470:3

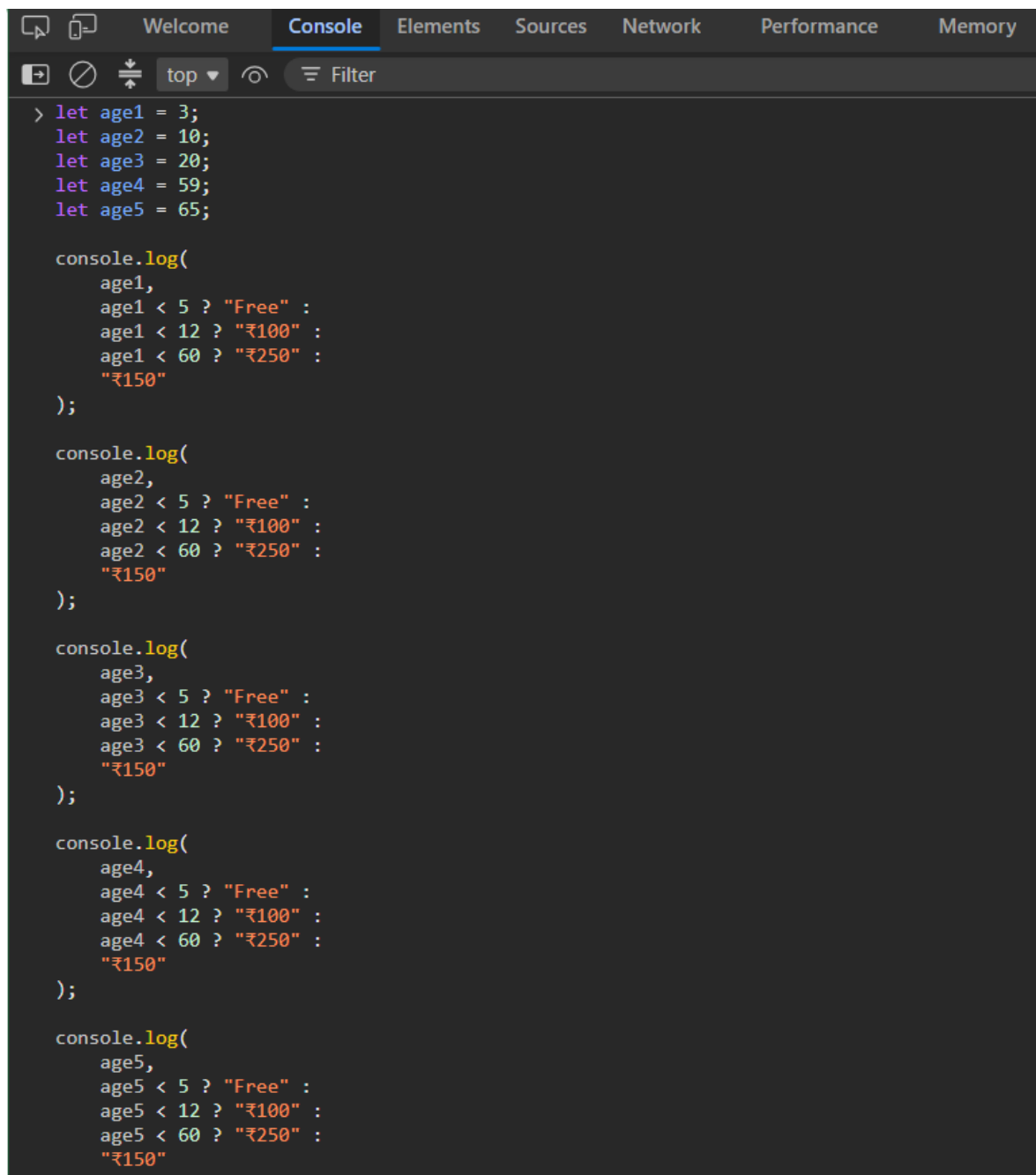
< undefined
>
```

Part 2 — Nested Ternary Decision Logic

Task 2.1 — Movie Ticket Pricing:

Task 2.2 — Shipping Cost Calculator:

Task 2.3 — Trickier: Ternary + Logical Operator Together:



```
> let age1 = 3;
    let age2 = 10;
    let age3 = 20;
    let age4 = 59;
    let age5 = 65;

    console.log(
      age1,
      age1 < 5 ? "Free" :
      age1 < 12 ? "₹100" :
      age1 < 60 ? "₹250" :
      "₹150"
    );

    console.log(
      age2,
      age2 < 5 ? "Free" :
      age2 < 12 ? "₹100" :
      age2 < 60 ? "₹250" :
      "₹150"
    );

    console.log(
      age3,
      age3 < 5 ? "Free" :
      age3 < 12 ? "₹100" :
      age3 < 60 ? "₹250" :
      "₹150"
    );

    console.log(
      age4,
      age4 < 5 ? "Free" :
      age4 < 12 ? "₹100" :
      age4 < 60 ? "₹250" :
      "₹150"
    );

    console.log(
      age5,
      age5 < 5 ? "Free" :
      age5 < 12 ? "₹100" :
      age5 < 60 ? "₹250" :
      "₹150"
    );
```

```

);

console.log(
  age2,
  age2 < 5 ? "Free" :
  age2 < 12 ? "₹100" :
  age2 < 60 ? "₹250" :
  "₹150"
);

console.log(
  age3,
  age3 < 5 ? "Free" :
  age3 < 12 ? "₹100" :
  age3 < 60 ? "₹250" :
  "₹150"
);

console.log(
  age4,
  age4 < 5 ? "Free" :
  age4 < 12 ? "₹100" :
  age4 < 60 ? "₹250" :
  "₹150"
);

console.log(
  age5,
  age5 < 5 ? "Free" :
  age5 < 12 ? "₹100" :
  age5 < 60 ? "₹250" :
  "₹150"
);

```

3 'Free'

10 '₹100'

20 '₹250'

59 '₹250'

65 '₹150'

← undefined

>

```

> let cartTotal = 1500;

let shipping =
  cartTotal >= 2000 ? "Free Shipping" :
  cartTotal >= 1000 ? "₹50" :
  "₹100";

console.log("Cart Total:", cartTotal);
console.log("Shipping:", shipping);

```

Cart Total: 1500

[VM440:8](#)

Shipping: ₹50

[VM440:9](#)

← undefined

>

```

> let cartTotal = 1500;

let shipping =
  cartTotal >= 2000 ? "Free Shipping" :
  cartTotal >= 1000 ? "₹50" :
  "₹100";

console.log("Cart Total:", cartTotal);
console.log("Shipping:", shipping);

Cart Total: 1500 VM440:8
Shipping: ₹50 VM440:9
< undefined
> let cart1 = 500;
let cart2 = 1500;
let cart3 = 2500;

console.log(
  cart1,
  cart1 >= 2000 ? "Free Shipping" :
  cart1 >= 1000 ? "₹50" :
  "₹100"
);

console.log(
  cart2,
  cart2 >= 2000 ? "Free Shipping" :
  cart2 >= 1000 ? "₹50" :
  "₹100"
);

console.log(
  cart3,
  cart3 >= 2000 ? "Free Shipping" :
  cart3 >= 1000 ? "₹50" :
  "₹100"
);

500 '₹100' VM444:5
1500 '₹50' VM444:12
2500 'Free Shipping' VM444:19
< undefined

```

```

> // Task 2.3 - Discount Type

let isMember = true;
let totalSpent = 6000;

let discountType =
  isMember && totalSpent >= 5000 ? "VIP Discount" :
  totalSpent >= 5000 ? "Regular Discount" :
  "No Discount";

console.log("Member:", isMember);
console.log("Total Spent:", totalSpent);
console.log("Discount Type:", discountType);

Member: true VM494:11
Total Spent: 6000 VM494:12
Discount Type: VIP Discount VM494:13
< undefined
> |

```

Part 3 — Manual Aggregation: Arrays & Objects without Loops

Task 3.1

Task 3.2

```
// Task 3.1 - Exam Scores

const scores = [60, 70, 80, 50, 90];

const totalScores =
  scores[0] +
  scores[1] +
  scores[2] +
  scores[3] +
  scores[4];

const averageScore = totalScores / scores.length;

const result = averageScore >= 40 ? "Pass" : "Fail";

console.log("----- TASK 3.1 -----");
console.log("Scores:", scores);
console.log("Total:", totalScores);
console.log("Average:", averageScore);
console.log("Result:", result);
```

----- TASK 3.1 -----	VM506:20
Scores: ► (5) [60, 70, 80, 50, 90]	VM506:21
Total: 350	VM506:22
Average: 70	VM506:23
Result: Pass	VM506:24

```
< undefined
>
```

```
> // Task 3.2 - Object Aggregation

const item1 = {
  name: "Notebook",
  price: 60,
  qty: 3
};

const item2 = {
  name: "Pen",
  price: 10,
  qty: 5
};

const item3 = {
  name: "Bag",
  price: 800,
  qty: 1
};

const item4 = {
  name: "Bottle",
  price: 200,
  qty: 2
};

< undefined
> // Task 3.2 - Object Aggregation

const item1 = {
  name: "Notebook",
  price: 60,
  qty: 3
};

const item2 = {
  name: "Pen",
  price: 10,
  qty: 5
};

const item3 = {
  name: "Bag",
  price: 800,
  qty: 1
};
```

```
const item3 = {
  name: "Bag",
  price: 800,
  qty: 1
};

const item4 = {
  name: "Bottle",
  price: 200,
  qty: 2
};

const subtotal1 = item1.price * item1.qty;
const subtotal2 = item2.price * item2.qty;
const subtotal3 = item3.price * item3.qty;
const subtotal4 = item4.price * item4.qty;

const grandTotal =
  subtotal1 +
  subtotal2 +
  subtotal3 +
  subtotal4;

console.log("----- TASK 3.2 -----");

console.log(item1.name, "Subtotal:", subtotal1);
console.log(item2.name, "Subtotal:", subtotal2);
console.log(item3.name, "Subtotal:", subtotal3);
console.log(item4.name, "Subtotal:", subtotal4);

console.log("Grand Total:", grandTotal);
```

----- TASK 3.2 -----

Notebook Subtotal: 180

Pen Subtotal: 50

Bag Subtotal: 800

Bottle Subtotal: 400

Grand Total: 1430

⏪ undefined

>

Part 4 — Deep Dive: Tricky Type Coercion

Answer - The most surprising result for me was `NaN === NaN`, which returns false. I expected two NaN values to be equal, but JavaScript treats NaN as a special value that is not equal to itself.

Task 4.2

```
> console.log(typeof NaN);  
number VM519:1  
◀ undefined
```

Answer- NaN means "Not-a-Number", but its JavaScript data type is still number. It represents an invalid or unsuccessful numeric result, rather than being a separate data type.

Part 5 — Mini Project: Smart Shopping Cart & Billing System

```
const cartItem1 = {
  name: "Notebook",
  price: 60,
  qty: 3
};

const cartItem2 = {
  name: "Pen",
  price: 10,
  qty: 5
};

const cartItem3 = {
  name: "Bag",
  price: 800,
  qty: 1
};

// Step 2: Defensive type check

console.log("Price 1 type:", typeof cartItem1.price);
console.log("Price 2 type:", typeof cartItem2.price);
console.log("Price 3 type:", typeof cartItem3.price);

const pricesAreNumbers =
  typeof cartItem1.price === "number" &&
  typeof cartItem2.price === "number" &&
  typeof cartItem3.price === "number";

console.log("All prices are numbers:", pricesAreNumbers);

// Step 3: Calculate subtotals and grand total

let cartSubtotal1;
let cartSubtotal2;
let cartSubtotal3;
let cartGrandTotal;

if (pricesAreNumbers) {
  cartSubtotal1 = cartItem1.price * cartItem1.qty;
  cartSubtotal2 = cartItem2.price * cartItem2.qty;
```

```
if (pricesAreNumbers) {

    cartSubtotal1 = cartItem1.price * cartItem1.qty;
    cartSubtotal2 = cartItem2.price * cartItem2.qty;
    cartSubtotal3 = cartItem3.price * cartItem3.qty;

    cartGrandTotal =
        cartSubtotal1 +
        cartSubtotal2 +
        cartSubtotal3;
}

// Step 4: Tiered discount

const discountPercent =
    cartGrandTotal >= 5000 ? 20 :
    cartGrandTotal >= 2000 ? 10 :
    cartGrandTotal >= 1000 ? 5 :
    0;

// Step 5: Discount amount

const discountAmount =
    cartGrandTotal * discountPercent / 100;

// Step 6: Amount after discount

const amountAfterDiscount =
    cartGrandTotal - discountAmount;

// Step 7: 18% GST

const gstAmount =
    amountAfterDiscount * 18 / 100;

// Step 8: Final payable amount

const finalPayable =
    amountAfterDiscount + gstAmount;
```

```

const finalPayable =
    amountAfterDiscount + gstAmount;

// Step 9: Free shipping using logical OR
const shippingStatus =
    amountAfterDiscount >= 1500 || 3 >= 3
    ? "FREE"
    : "₹100 shipping";

// Step 10: Print receipt
console.log(`
===== SHOPPING RECEIPT =====

${cartItem1.name} - ₹${cartItem1.price} × ${cartItem1.qty}
Subtotal: ₹${cartSubtotal1}

${cartItem2.name} - ₹${cartItem2.price} × ${cartItem2.qty}
Subtotal: ₹${cartSubtotal2}

${cartItem3.name} - ₹${cartItem3.price} × ${cartItem3.qty}
Subtotal: ₹${cartSubtotal3}

-----

Grand Total: ₹${cartGrandTotal}

Discount: ${discountPercent}%
Discount Amount: ₹${discountAmount.toFixed(2)}

Amount After Discount: ₹${amountAfterDiscount.toFixed(2)}

GST (18%): ₹${gstAmount.toFixed(2)}

Final Payable: ₹${finalPayable.toFixed(2)}

Shipping: ${shippingStatus}

=====
`);

```

```

Final Payable: ₹${finalPayable.toFixed(2)}
Shipping: ${shippingStatus}
=====
`);

```

```

Price 1 type: number
Price 2 type: number
Price 3 type: number

```

```

All prices are numbers: true

```

```

===== SHOPPING RECEIPT =====

```

```

Notebook - ₹60 × 3
Subtotal: ₹180

```

```

Pen - ₹10 × 5
Subtotal: ₹50

```

```

Bag - ₹800 × 1
Subtotal: ₹800

```

```

-----

Grand Total: ₹1030

```

```

Discount: 5%
Discount Amount: ₹51.50

```

```

Amount After Discount: ₹978.50

```

```

GST (18%): ₹176.13

```

```

Final Payable: ₹1154.63

```

```

Shipping: FREE

```

```

=====

```

```

< undefined

```

```

>

```

Objective

To create a simple shopping cart and billing system using JavaScript objects, calculations, ternary operators and logical operators.

Working / Explanation

- First, three products were created using objects with their name, price and quantity.
- The subtotal of each product was calculated using price × quantity.
- All subtotals were added manually to get the Grand Total.
- A nested ternary operator was used to calculate the discount according to the total amount.
- 18% GST was calculated on the amount after discount.
- The logical OR (||) operator was used to determine free shipping.
- typeof was used to check that all product prices were Numbers.
- Finally, all details were displayed as a shopping receipt in the console.

Result

The program successfully calculated the subtotal, grand total, discount, GST, final payable amount and shipping status.

Then directly apna Console ka screenshot paste kar do.

Ye documentation ke liye short bhi hai aur teacher ko clearly samajh aa jayega ki tumne kya-kya use kiya hai.

Part 6 — Debugging Challenge: Multi-Bug Snippets

Snippet 1

```
> let price = 500;
   let discount = price * 0.1;
   discount = discount + 5;
   console.log("Final: " + discount);

Final: 55 VM632:4
< undefined
```

Bug 1: const variable was reassigned, so it was changed to let.

Bug 2: fnal was an incorrect variable name, so it was corrected to discount

Snippet 2 (silent bug — no crash on the first bug!)

```
let studentMarks = 85;

let passResult =
  studentMarks === 85 ? "Pass" : "Fail";

console.log(passResult);

Pass VM668:8
< undefined
> |
```

Snippet 3

```
> let cartTotal = 1200;

   cartTotal = cartTotal - 100;

   let shipping =
     cartTotal >= 1500 ? "Free" : "Paid";

   console.log(shipping);

Paid VM664:8
< undefined
```

Thank you